

RATNAGIRI, India

WMO: 431100

Lat: 16.983N

Lon: 73.333E

Elev: 67

StdP: 100.52

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.9	18.9	6.7	6.1	27.0	8.8	7.1	26.6	6.4	27.3	5.4	27.2	1.0	90	0.301

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	5.8	34.4	23.2	33.6	23.6	33.0	23.8	27.6	30.9	27.2	30.5	26.9	30.2	2.2	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.7	22.4	29.4	26.3	21.9	29.1	26.0	21.6	28.8	88.1	31.0	86.6	30.6	85.2	30.3	30.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.6	5.0	4.2	DB	15.6	37.5	1.5	1.8	14.5	38.8	13.7	39.9	12.9	40.9	11.8	42.1	(4)
(5)			WB	11.7	28.3	1.1	0.7	10.9	28.8	10.3	29.2	9.7	29.6	8.9	30.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec			
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)			
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	27.1	25.1	25.6	27.0	28.5	29.6	27.7	26.7	26.6	27.9	27.7	26.3	(6)		
(7)		DBStd	1.79	1.61	1.70	1.46	1.08	1.00	1.69	1.11	0.87	0.95	1.31	1.26	1.52	(7)	
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
(9)		HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)	
(10)		CDD10.0	6253	468	436	528	556	608	531	519	514	499	556	531	506	(10)	
(11)		CDD18.3	3212	210	203	269	306	350	281	261	256	249	297	281	247	(11)	
(12)		CDH23.3	33177	1798	1815	2762	3715	4626	3089	2492	2337	2247	3111	2920	2266	(12)	
(13)	CDH26.7	11885	679	706	1056	1525	2198	1040	455	339	481	1195	1275	937	(13)		
(14)	Wind	WSAvg	1.6	1.4	1.5	1.4	1.6	1.8	2.2	1.9	1.2	1.1	1.3	1.5	(14)		
(15)	Precipitation	PrecAvg	2851	1	0	0	2	30	817	934	598	348	89	43	11	(15)	
(16)		PrecMax	3923	16	2	5	13	181	1319	1418	1358	828	427	295	144	(16)	
(17)		PrecMin	1836	0	0	0	0	0	382	448	133	117	0	0	0	(17)	
(18)		PrecStd	580	3	0	1	3	47	235	246	329	195	102	69	33	(18)	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.6	34.7	35.2	34.4	34.4	33.6	30.4	30.1	31.7	35.1	34.9	34.0	(19)	
MCWB			20.3	19.7	22.1	24.6	26.1	26.5	26.5	26.0	26.2	23.4	22.4	21.2	(20)		
2%		DB	32.4	33.0	33.0	32.9	33.5	32.3	29.7	29.3	30.5	34.0	33.9	32.9	(21)		
		MCWB	20.3	20.8	23.1	25.3	26.4	26.4	26.3	25.9	25.7	23.9	22.5	21.1	(22)		
5%		DB	31.4	31.8	31.9	32.2	32.8	31.5	29.2	28.8	29.7	32.9	33.1	32.1	(23)		
		MCWB	20.2	21.2	23.6	25.4	26.2	26.4	26.1	25.8	25.5	24.5	22.5	20.8	(24)		
10%		DB	30.3	30.6	31.0	31.6	32.3	30.6	28.7	28.4	29.1	31.7	32.3	31.2	(25)		
		MCWB	20.4	21.5	24.0	25.3	26.0	26.4	26.0	25.6	25.4	25.0	22.6	20.8	(26)		
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.0	25.1	26.7	27.7	28.2	27.9	27.2	26.9	27.2	27.7	26.5	25.2	(27)	
MCDWB			28.6	29.1	30.8	31.2	31.9	30.8	29.2	28.9	29.9	31.2	30.7	29.7	(28)		
2%		WB	23.1	24.2	26.2	27.1	27.6	27.4	26.7	26.4	26.6	27.1	25.8	24.3	(29)		
		MCDWB	28.3	28.7	30.1	30.8	31.4	30.3	28.8	28.3	29.0	30.5	29.9	29.2	(30)		
5%		WB	22.5	23.5	25.7	26.6	27.2	27.1	26.5	26.1	26.2	26.6	25.3	23.6	(31)		
		MCDWB	28.0	28.3	29.5	30.4	31.1	30.0	28.5	28.1	28.5	30.0	29.5	28.7	(32)		
10%		WB	21.9	22.8	25.2	26.2	26.8	26.7	26.2	25.8	25.8	26.1	24.8	22.8	(33)		
		MCDWB	27.7	27.9	29.0	30.1	30.8	29.5	28.1	27.7	28.1	29.4	29.1	28.0	(34)		
(35)	Mean Daily Temperature Range	5% DB	MDBR	11.0	10.6	8.7	6.7	5.8	4.4	3.1	3.2	4.7	7.7	10.2	11.0	(35)	
MCDBR			12.1	12.1	9.8	7.3	6.3	5.7	3.8	3.8	5.8	10.0	11.2	11.7	(36)		
MCWBR			5.6	5.6	4.6	2.9	2.1	2.0	1.9	1.8	2.4	3.9	4.7	5.2	(37)		
5% WB		MCDBR	10.9	10.4	7.6	6.3	6.0	4.8	3.5	3.5	4.8	7.3	8.6	10.0	(38)		
		MCWBR	5.2	5.1	3.8	2.9	2.5	2.4	2.0	1.9	2.4	3.4	3.5	4.4	(39)		
(40)	Clear-Sky Solar Irradiance	taub	0.456	0.459	0.535	0.592	0.570	0.631	0.667	0.608	0.537	0.587	0.528	0.476	(40)		
taud		2.153	2.167	1.984	1.879	1.961	1.840	1.755	1.886	2.080	1.905	2.024	2.114	(41)			
Ebn at Noon		825	845	792	748	754	702	679	726	780	724	751	792	(42)			
Edh at Noon		145	150	185	206	188	210	229	202	166	192	164	147	(43)			
(44)	All-Sky Solar Radiation	RadAvg	5.23	6.03	6.69	7.08	6.78	4.27	3.50	3.90	4.68	5.17	5.06	4.94	(44)		
(45)		RadStd	0.16	0.14	0.20	0.17	0.39	0.50	0.54	0.31	0.45	0.28	0.26	0.13	(45)		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.15	N/A	+1.05	+0.41	+0.34	+0.37	N/A	N/A	N/A	N/A
(47)	Regional (3 neighbors)	+0.19	N/A	+0.58	+0.40	+0.29	+0.27	N/A	N/A	N/A	N/A

Nomenclature: See separate page